

下面这份是按你上传的 **HTML slides** 当前 **12** 页顺序 写的完整 presentation script。Deck 里包含标题页、Problem & Design Idea、System Architecture、Project Workflow、Dataset Strategy、Preprocessing Pipeline、Specialist Models、Training Protocol、Model Evidence Results、Temporal Decision Fusion、Working System Interface 和 Closing Demo 页。

建议总时长控制在 **7** 分钟左右。每页不要逐字读，只背每页的核心点，然后用英文自然讲出来。

Page 1 — VisionGuard Title

Time: 25–30 seconds

English Script

Good morning everyone. Today we are presenting **VisionGuard**, an AI-powered driver drowsiness monitoring prototype.

The main idea is to turn visible evidence, such as eye closure and yawning, into transparent warning candidates.

This project is not just a model training experiment. It connects data preprocessing, specialist CNN models, temporal fusion, and a working web interface for both real-time monitoring and uploaded-video analysis.

中文脚本

大家好，今天我们展示的项目是 **VisionGuard**，一个基于 AI 的驾驶员疲劳监测原型系统。它的核心思想是把可见证据，例如闭眼和打哈欠，转化成透明、可解释的预警候选状态。这个项目不只是一个模型训练实验，而是把数据预处理、专门 CNN 模型、时间融合逻辑，以及实时监测和视频上传分析界面连接成一个完整系统。

核心背诵点 / Key Points

EN

- VisionGuard is a modular drowsiness monitoring prototype.
- It turns eye and yawn evidence into warning candidates.
- The project includes models, fusion logic, and a working web system.

中文

- VisionGuard 是模块化疲劳监测原型。
- 它把眼部和打哈欠证据转化为预警候选状态。

- 项目包含模型、融合逻辑和实际 Web 系统。

Page 2 — Problem & Design Idea

Time: 40 seconds

English Script

The key problem is that drowsiness is not a single-frame visual label.

One closed-eye frame may simply be a blink, and one mouth movement may not prove yawning. Also, if the face is lost or the driver turns away, that should be treated as signal quality, not as drowsiness.

So our design avoids a single black-box “drowsy or not-drowsy” output. Instead, VisionGuard uses modular eye and mouth specialists, then fuses their evidence over time into explainable warning states.

中文脚本

这个项目的核心问题是：疲劳不是一个单帧图像标签。

一帧闭眼可能只是眨眼，一次嘴部动作也不一定能证明打哈欠。同时，如果人脸丢失或者驾驶员转头，这应该被看作信号质量问题，而不是疲劳证据。

所以我们的设计没有使用一个黑箱式的“疲劳或不疲劳”输出，而是使用眼部和嘴部两个模块，再通过时间融合生成可解释的预警状态。

核心背诵点 / Key Points

EN

- Drowsiness needs evidence over time, not one snapshot.
- Lost-face cases are signal-quality issues.
- Our design is modular: eye specialist + mouth specialist + temporal fusion.

中文

- 疲劳需要时间证据，不是单帧判断。
- 人脸丢失是信号质量问题，不是疲劳证据。
- 系统设计是眼部模块 + 嘴部模块 + 时间融合。

Page 3 — System Architecture

Time: 40 seconds

English Script

This slide shows the system architecture.

The user interacts with the system through Live Monitor, Video Upload Analysis, History, and Insights. The input layer handles webcam or uploaded video, frame sampling, face detection, and eye and mouth ROI extraction.

Then two specialist CNNs generate two evidence signals: `p_eye_closed` from the eye model and `p_yawn` from the mouth model.

Finally, the temporal fusion layer combines those signals with signal-quality checks and produces warning states for the user interface and history records.

中文脚本

这一页展示系统架构。

用户通过 Live Monitor、Video Upload Analysis、History 和 Insights 使用系统。输入层负责处理摄像头或上传视频，包括帧采样、人脸检测，以及眼睛和嘴部 ROI 提取。

然后两个专门 CNN 分别生成两个证据信号：眼部模型输出 `p_eye_closed`，嘴部模型输出 `p_yawn`。

最后，时间融合层结合这些信号和信号质量检查，生成预警状态，并传递给界面和历史记录模块。

核心背诵点 / Key Points

EN

- UI receives webcam or uploaded video input.
- Eye model outputs `p_eye_closed`; mouth model outputs `p_yawn`.
- Fusion layer converts evidence into warning states.

中文

- 系统输入来自摄像头或上传视频。
- 眼部模型输出 `p_eye_closed`，嘴部模型输出 `p_yawn`。
- 融合层把证据转化为预警状态。

Page 4 — Project Workflow

Time: 35 seconds

English Script

Our workflow has two clear phases: offline training and online runtime.

In the offline phase, we validate datasets, prepare crops, split data by subject, train specialist models, and select the best checkpoints.

In the runtime phase, the system captures frames from a webcam or uploaded video, extracts eye and mouth regions, runs the selected CNNs, applies temporal fusion, and then presents alerts, keyframes, history, and insights.

This separation makes the system easier to explain and maintain.

中文脚本

我们的工作流程分为两个阶段：离线训练和在线运行。

离线阶段包括数据验证、裁剪图生成、按 subject 划分数据、训练专门模型，以及选择最佳 checkpoint。

在线运行阶段中，系统从摄像头或上传视频中获取帧，提取眼部和嘴部区域，运行选定的 CNN 模型，再进行时间融合，最后展示预警、关键帧、历史记录和分析洞察。

这种分离让系统更容易解释，也更容易维护。

核心背诵点 / Key Points

EN

- Offline training prepares models and checkpoints.
- Runtime only loads selected models and performs inference.
- The final product connects inference, alerts, history, and insights.

中文

- 离线阶段负责训练模型和选择 checkpoint。
- 运行阶段只加载模型并进行推理。
- 最终系统连接了推理、预警、历史和分析页面。

Page 5 — Dataset Strategy

Time: 45 seconds

English Script

We used two complementary datasets because the system has two specialist tasks.

For eye open and closed detection, we used the MRL Eye corpus, with 84,898 images from 37

subjects.

For mouth and yawn detection, we used YawDD and YawDD+ Dash data, producing 64,202 mouth crops from 29 drivers or subjects.

The most important design choice here is subject-level splitting. No subject appears across train, validation, and test sets, which helps prevent identity leakage and frame leakage.

These are specialist labels, supporting a modular design rather than one end-to-end drowsiness label.

中文脚本

我们使用两个互补数据集，因为系统有两个专门任务。

眼部开闭检测使用 MRL Eye 数据集，包括 84,898 张图像，来自 37 个 subject。

嘴部和打哈欠检测使用 YawDD 和 YawDD+ Dash 数据，生成了 64,202 张嘴部裁剪图，来自 29 个 driver 或 subject。

这一页最重要的设计是 subject-level split。同一个 subject 不会同时出现在训练集、验证集和测试集中，这有助于防止 identity leakage 和 frame leakage。

这些标签是专门模块标签，服务于模块化设计，而不是单一的端到端疲劳标签。

核心背诵点 / Key Points

EN

- Two datasets support two specialist evidence streams.
- MRL Eye is for eye open/closed; YawDD is for mouth/yawn.
- Subject-level split prevents leakage.

中文

- 两个数据集对应两个专门证据流。
- MRL Eye 用于眼部开闭，YawDD 用于嘴部打哈欠。
- subject-level split 防止数据泄漏。

Page 6 — Preprocessing Pipeline

Time: 40 seconds

English Script

This slide explains how we prepared the YawDD mouth training data.

The input includes raw YawDD Dash videos and YawDD+ frame annotations. First, we reconstruct labelled frames from the raw videos. Then, we use MediaPipe lip landmarks to crop the mouth ROI.

If landmarks fail, the system tries a lower-face fallback, and unusable crops are excluded. Finally, the data is split by driver or subject. The output is 64,202 training-ready mouth crops, with a 99.73% crop success rate. This pipeline is only for YawDD mouth data. Runtime ROI extraction is separate.

中文脚本

这一页解释 YawDD 嘴部训练数据是如何准备的。

输入包括原始 YawDD Dash 视频和 YawDD+ 帧标注。第一步是根据标注从原始视频中重建带标签的帧。然后，系统使用 MediaPipe 嘴唇关键点裁剪嘴部 ROI。

如果关键点失败，系统会尝试下半脸 fallback，无法使用的 crop 会被排除。

最后，数据按 driver 或 subject 划分。最终输出是 64,202 张可用于训练的嘴部 crop，crop 成功率为 99.73%。

这个流程只用于 YawDD 嘴部训练数据，运行时的 ROI 提取是单独流程。

核心背诵点 / Key Points

EN

- YawDD+ annotations guide frame reconstruction.
- MediaPipe lip landmarks generate mouth crops.
- The output is 64,202 trainable mouth crops with 99.73% crop success.

中文

- YawDD+ 标注用于重建带标签帧。
- MediaPipe 嘴唇关键点用于生成嘴部 crop。
- 最终得到 64,202 张训练 crop，成功率 99.73%。

Page 7 — Specialist Models

Time: 45 seconds

English Script

This slide shows the three core components of VisionGuard.

The first component is the eye specialist: MobileNetV2 trained on MRL Eye. It outputs `p_eye_closed` and achieved 98.6% specialist test accuracy.

The second component is the mouth specialist: ResNet18 trained on YawDD mouth crops. It outputs `p_yawn` and achieved 99.4% specialist test accuracy.

The third component is not another CNN. It is a temporal rule layer that combines `p_eye_closed`, `p_yawn`, and temporal persistence into explainable warning states.

中文脚本

这一页展示 VisionGuard 的三个核心组件。

第一个是眼部专门模型：在 MRL Eye 上训练的 MobileNetV2。它输出 `p_eye_closed`，专门测试准确率为 98.6%。

第二个是嘴部专门模型：在 YawDD 嘴部 crop 上训练的 ResNet18。它输出 `p_yawn`，专门测试准确率为 99.4%。

第三个组件不是另一个 CNN，而是时间规则层。它把 `p_eye_closed`、`p_yawn` 和时间持续性融合成可解释的预警状态。

核心背诵点 / Key Points

EN

- MobileNetV2 is selected for eye open/closed detection.
- ResNet18 is selected for mouth/yawn detection.
- Fusion is rule-based, not a trained fusion model.

中文

- MobileNetV2 是眼部模块的最终模型。
- ResNet18 是嘴部模块的最终模型。
- 融合层是规则逻辑，不是训练出来的融合模型。

Page 8 — Training Protocol

Time: 35 seconds

English Script

For both specialist tasks, we used transfer learning with ImageNet-pretrained CNN backbones, including ResNet18, MobileNetV2, and EfficientNet-B0.

The protocol has two stages. First, we replace the classifier head and train only the new head while the backbone is frozen. Second, we unfreeze the backbone and fine-tune the full model.

We also used 224 by 224 RGB inputs, ImageNet normalization, weighted CrossEntropyLoss, and subject-level train, validation, and test splits to keep evaluation leakage-controlled.

中文脚本

两个专门任务都使用了基于 ImageNet 预训练 CNN backbone 的迁移学习，包括 ResNet18、MobileNetV2 和 EfficientNet-B0。

训练流程分为两个阶段。第一阶段替换分类头，并在冻结 backbone 的情况下只训练新的分类头。第二阶段解冻 backbone，对整个模型进行 fine-tuning。

我们还使用 224×224 RGB 输入、ImageNet normalization、weighted CrossEntropyLoss，以及 subject-level 的训练、验证和测试划分，以控制评估泄漏。

核心背诵点 / Key Points

EN

- We used ImageNet-pretrained CNN backbones.
- Training has two stages: frozen head training, then full fine-tuning.
- Subject-level split keeps evaluation leakage-controlled.

中文

- 使用 ImageNet 预训练 CNN backbone。
- 两阶段训练：先冻结训练分类头，再整体 fine-tune。
- subject-level split 控制评估泄漏。

Page 9 — Model Evidence Results

Time: 60 seconds

English Script

This slide shows how we selected the runtime models. The key point is that the comparison is within each specialist task, not one unified drowsiness benchmark.

For the mouth and yawn specialist, ResNet18 achieved the best result among the three CNN baselines, with 99.4% test accuracy, 97.2% yawn F1, and 97.9% yawn recall. So ResNet18 was selected for `p_yawn`.

For the eye open and closed specialist, MobileNetV2 achieved the best default balance, with 98.6% test accuracy, 98.6% macro F1, and 98.5% closed-eye recall. So MobileNetV2 was selected for `p_eye_closed`.

The final tab summarizes the design: specialist metrics support module selection, and temporal fusion produces the final warning states. The HTML deck explicitly separates the mouth, eye, and final selection tabs.

中文脚本

这一页展示我们如何选择最终运行模型。关键点是，这里的比较是在每个专门任务内部进行的，不是一个统一的疲劳检测 benchmark。

对于嘴部和打哈欠模块，ResNet18 在三个 CNN baseline 中表现最好，测试准确率为 99.4%，yawn F1 为 97.2%，yawn recall 为 97.9%。因此 ResNet18 被选为 `p_yawn` 模型。

对于眼部开闭模块，MobileNetV2 在默认设置下取得最好的综合平衡，测试准确率为 98.6%，macro F1 为 98.6%，closed-eye recall 为 98.5%。因此 MobileNetV2 被选为 `p_eye_closed` 模型。

最后一个 tab 总结了系统设计：专门指标支持模块选择，而最终预警状态由时间融合产生。

核心背诵点 / Key Points

EN

- The benchmarks are within-task specialist comparisons.
- Mouth specialist selected ResNet18.
- Eye specialist selected MobileNetV2.
- Final warning states come from temporal fusion, not accuracy ranking alone.

中文

- 这是任务内部的专门模型比较。
- 嘴部模块选择 ResNet18。
- 眼部模块选择 MobileNetV2。
- 最终预警状态来自时间融合，不只是 accuracy 排名。

Page 10 — Temporal Decision Fusion

Time: 60 seconds

English Script

This slide explains how warning states emerge over time.

The eye row shows rolling `p_eye_closed`. The system does not react to one isolated frame. It looks for sustained eye-warning evidence.

The mouth row shows `p_yawn` events and recent-yawn context. A yawn alone is not treated as a final verdict, but it becomes useful evidence for fusion.

The signal-quality row handles cases where the face is lost. Those cases become

`signal_unreliable`, not drowsiness evidence.

Finally, the fusion state changes from normal to mouth-warning and then to high-confidence

candidate when sustained eye evidence overlaps with recent yawn evidence. The controlled clip evidence shows 103 sampled frames, 18 eye-warning frames, 30 mouth-warning frames, and 6 high-confidence overlap frames.

中文脚本

这一页解释预警状态如何随着时间产生。

眼部这一行展示 rolling `p_eye_closed`。系统不会因为单独一帧就立刻反应，而是寻找持续的眼部预警证据。

嘴部这一行展示 `p_yawn` 事件和 recent-yawn context。一次打哈欠本身不是最终结论，但它会成为融合逻辑中的重要证据。

信号质量这一行处理人脸丢失的情况。这些情况会被标记为 `signal_unreliable`，而不是疲劳证据。

最后，融合状态会从 normal 变成 mouth-warning，并在持续眼部证据和近期打哈欠证据重叠时变成 high-confidence candidate。控制视频的结果显示：103 个采样帧、18 个 eye-warning 帧、30 个 mouth-warning 帧，以及 6 个 high-confidence overlap 帧。

核心背诵点 / Key Points

EN

- Warning states emerge over time, not from one frame.
- High confidence requires sustained eye evidence plus recent yawn.
- Lost face is signal quality, not drowsiness evidence.
- Controlled clip evidence supports the rule behavior.

中文

- 预警状态来自时间证据，不是单帧判断。
- 高置信候选需要持续眼部证据 + 近期打哈欠。
- 人脸丢失是信号质量问题，不是疲劳证据。
- 控制视频结果支持规则逻辑。

Page 11 — Working System Interface

Time: 45 seconds

English Script

This slide shows the working VisionGuard interface.

The system includes four major views: Live Monitor, Video Upload Analysis, History, and Insights. In Live Monitor, the user can see the camera feed, eye and yawn event counts, the Drowsiness Risk gauge, the waveform chart, and recent events.

The important point is that the interface is not only for displaying a prediction. It helps the user understand what evidence contributed to the warning and supports later review through history and insights. The HTML slide also allows switching between these interface tabs.

中文脚本

这一页展示 VisionGuard 的实际系统界面。

系统包括四个主要视图：Live Monitor、Video Upload Analysis、History 和 Insights。在 Live Monitor 中，用户可以看到摄像头画面、闭眼和打哈欠事件计数、Drowsiness Risk 仪表盘、波形图和最近事件。

重点是，这个界面不只是显示一个预测结果，而是帮助用户理解预警背后的证据，并通过 History 和 Insights 支持后续复查。HTML slide 中也可以切换这些界面 tab。

核心背诵点 / Key Points

EN

- The system has Live Monitor, Video Upload, History, and Insights.
- The UI shows evidence, risk, events, and review history.
- It is a working prototype, not just a notebook result.

中文

- 系统包括实时监测、视频上传、历史和洞察页面。
- UI 展示证据、风险、事件和复查记录。
- 这是实际可运行原型，不只是 notebook 结果。

Page 12 — Closing / Demo

Time: 25–30 seconds

English Script

To conclude, VisionGuard moves from visual evidence to driver awareness.

It uses specialist models to detect eye and mouth cues, temporal fusion to combine those cues over time, and a web interface to present explainable warning states.

In the live demo, we will start the camera, show warning behavior, upload a clip for interval and keyframe review, and then open History and Insights to show the evidence trail.

中文脚本

最后总结一下，VisionGuard 的核心是从视觉证据走向驾驶员意识提醒。
它使用专门模型检测眼部和嘴部线索，使用时间融合把这些线索结合起来，并通过 Web 界面展示可解释的预警状态。
在 live demo 中，我们会启动摄像头，展示预警行为，上传视频来查看融合区间和关键帧，然后打开 History 和 Insights 展示证据轨迹。

核心背诵点 / Key Points

EN

- VisionGuard turns evidence into awareness.
- Specialist models detect cues; fusion creates warning states.
- Demo flow: camera, upload analysis, history, insights.

中文

- VisionGuard 把视觉证据转化为风险意识。
- 专门模型检测线索，融合层生成预警状态。
- Demo 流程：摄像头、上传分析、历史、洞察。

7-Minute Timing Plan

Page	Time
1 Title	0:25
2 Problem	0:40
3 Architecture	0:40
4 Workflow	0:35
5 Dataset	0:45
6 Preprocessing	0:40
7 Specialist Models	0:45
8 Training	0:35
9 Model Evidence	1:00
10 Fusion	1:00
11 Interface	0:45

12 Closing	0:25
Total	7:15

实际演讲时，把 Page 9 或 Page 10 各压缩 5–8 秒，就能稳定控制在 7 分钟内。

最重要的 8 句话，必须背熟

English

1. VisionGuard is a modular driver drowsiness monitoring prototype.
2. Drowsiness is not a single-frame problem.
3. We use two specialist evidence streams: `p_eye_closed` and `p_yawn`.
4. The eye specialist uses MobileNetV2 trained on MRL Eye.
5. The mouth specialist uses ResNet18 trained on YawDD mouth crops.
6. Subject-level splitting helps prevent identity and frame leakage.
7. Temporal fusion combines eye evidence, yawn evidence, persistence, and signal quality.
8. The working interface presents warnings, evidence, history, and insights.

中文

1. VisionGuard 是一个模块化驾驶员疲劳监测原型。
2. 疲劳不是单帧判断问题。
3. 系统使用两个专门证据流： `p_eye_closed` 和 `p_yawn`。
4. 眼部模块使用在 MRL Eye 上训练的 MobileNetV2。
5. 嘴部模块使用在 YawDD 嘴部 crop 上训练的 ResNet18。
6. subject-level split 有助于防止身份和帧泄漏。
7. 时间融合结合眼部证据、打哈欠证据、持续性和信号质量。
8. 实际界面展示预警、证据、历史记录和分析洞察。